

Case Study: How Netflix Is Using AWS and its Own CDN

[Ayush Kumar singh](#)



Netflix is one of the Biggest customer of AWS and it uses AWS for nearly all its computing and storage needs, including databases, analytics, recommendation engines, video transcoding, and more hundreds of functions that in total use more than 100,000 server instances on AWS.

Let See How Netflix Started Using AWS Services for growth and scalability.

How AWS Can Help Any Company or Startup?

Amazon Web Services (AWS) is the world's most comprehensive and broadly adopted cloud platform, offering over 175 fully-featured services from data centers globally.

For almost all Startups rapid growth and scalability are the keys to success. One of the biggest obstacles startups face is how to grow their customer base without overloading their servers. Everyone wants a million

subscribers. But not every new startup has the computing power to handle that much traffic, and it doesn't make economic sense to buy the extra bandwidth before you have the customer base to support it.

Here, AWS Can Help These Startups, It Provides on-demand access to all these sources and pay for them as you go. So whether you're a recent startup or a fully-funded business AWS has Your Back.

Why AWS?

1. Amazon's AWS has everything you need.
2. AWS Provides Scalability and Reliability
3. AWS is continuously innovating
4. AWS has Strong Monitoring tools.
5. AWS gives startups worry-free backups.
6. Access to a wide range of affordable services without contracts and legal terminologies.
7. Testing and development are easy on AWS and can create new development and testing environments on the fly.

Let's See How Netflix Is Using It.

Netflix Earning Model:

Netflix is a subscription-based business model making money with three simple plans: **Basic, Standard, and Premium**, giving access to stream series, movies, and shows. But it's not all it is also based on the concept of on-demand. It is a **media production company**. It is a brand that in the mind of its subscribers can mean several things. In fact, among the over a hundred thousand subscribers some tribes get assembled around the Netflix series which has become the symbol of our generation.

Netflix Without AWS: Back in 2000

Netflix, Inc. originally began its inception in **1998** by providing services to customers through means of mailing out physical copies(DVDs) of movies, shows, video games, and other forms of media through a standard mailing system.

Due to rapid changes in technology, Netflix converted its business model. They went from physical copies handouts to allowing customers streaming their favorite contents from the comfort of their own convenience.

In 2006, Amazon introduces video on demand service Amazon Prime Videos and Later that Year Netflix delivers its billionth DVD and begins to move away from its original core business model of mailing DVDs by introducing video on demand via the Internet. Their Amazon Becomes Netflix Biggest competitor.

In 2007, Netflix built two datacenters, located right next to each other but the company could not build data centers fast enough to meet the often-spiky demand of its users And as soon they got everything working they would run out of capacity, and the whole process had to start over again.

The experience of building datacenters taught Netflix an important lesson — they weren't good at building datacenters.

What Netflix was good at was delivering video to their members. Netflix would rather concentrate on getting better at delivering video rather than getting better at building datacenters.

Building Datacenters is not a competitive advantage for Netflix, delivering video is.

When and Why Netflix Decided to Move On AWS.

For three days in August 2008, Netflix could not ship DVDs because of corruption in their database. This was unacceptable. Netflix had to do

something.

At that time, Netflix decided to move to AWS even though Amazon is their biggest competitor. AWS was just getting established, so selecting AWS was a bold move.

Netflix moved to AWS because it wanted a more reliable infrastructure. Netflix wanted to remove any single point of failure from its system. AWS offered highly reliable databases, storage, and redundant data centers. Netflix wanted cloud computing, so it wouldn't have to build big unreliable monoliths anymore. Netflix wanted to become a global service without building its own datacenters. None of these capabilities were available in its old datacenters and never would be.

It took almost 8 years for Netflix to migrate to AWS. In early January 2016, Netflix completed their cloud migration and have shut down the last remaining data center bits used by their streaming service.

Get Ayush Kumar Singh's stories in your inbox

Join Medium for free to get updates from this writer.

But With the Increase No. of Netflix User They decided to Build their own CDN (Context Delivery Network) **Open Connect** (*Discussed later*) for Storing and Transmitting Data. And Rest Everything is Handled By Amazon Web Services

How AWS Helps Netflix?

Netflix is currently present in more than 190 countries. The elasticity of the cloud platform allowed them to add 1000s of virtual servers within a matter of a few minutes. They were able to expand their services to all the countries with minimal or rather negligible issues.

Leveraging multiple AWS cloud regions across geographies enabled

Netflix to dynamically shift and expand their global infrastructure capacity across all countries they are present in. This provided for a brilliant customer experience wherein entertainment provided by Netflix continued to rule, uninterrupted.

Amazon Kinesis Data Streams processes multiple terabytes of log data each day, yet events show up in our analytics in seconds. We can discover and respond to issues in real-time, ensuring high availability, and great customer experience.

-John Bennett

Senior Software Engineer, Netflix

Aws Helps Improving Customer Experience with Real-Time Network Monitoring. Netflix is now able to identify new ways to optimize its applications, whether that means moving an application from one region to another or changing to a more appropriate network protocol for a specific type of traffic. "Our solution built on Amazon Kinesis enables us to identify ways to increase efficiency, reduce costs, and improve resiliency for the best customer experience," says Bennett.

And The Most Important AWS is way cheaper than their personal cloud.

Note : Netflix Does not use AWS for Storing Content and Transmitting it. Then have Their OWN CDN.

What AWS Services Netflix Use And Why?

Netflix has a very complex architecture on AWS and They are Using all of them together as a Hybrid system to Stream and power their Services. Here I Tried Explaining Some of them Only.

1. Amazon Kinesis Data Streams (KDS)

KDS helps Netflix to continuously capture gigabytes of data per second

from hundreds of thousands of its sources and the data collected is available in milliseconds to enable real-time analytics use and Stored in **Amazon S3**. Which can be used for Improving Customer Experience and Their Recommender system.

2. AWS Lambda

With AWS Lambda Netflix build a rule-based self-managing infrastructure and replace inefficient processes to reduce the rate of errors and save valuable time.

With lambda, we can use the rules that trigger on the data updates that decide what needs to be backed up, what needs to be copied to off-site storage and to check and validate that it arrived safely and to restart the copies and recheck and revalidate if it didn't raise alarms in case of failure.

It is also used for Automating Security, lambda allows them to validate that each new instance is constructed and configured in accordance with the rules and situations and to trigger shutdown the violations or notification of unauthorized instances that appear in their infrastructure.

3. Amazon Route 53

Netflix is present in more than 190 countries. All use Different AWS Regions and to make it resilient for users, A complex DNS architecture is required this Is Where AWS Route 53 Helps Netflix.

In Case for Failure or Overloading in a particular Server or Region Route 53 helps Netflix to quickly and adequately route the traffic to different regions.

There are no more services that Netflix uses like EC2, ELB, TTL, machine learning, etc. AWS in some way or the other.

How Netflix Provides Seamless and Low Latency Video Streaming Around The Globe ??

Here, They Don't rely on AWS servers.

In Jan. 2016 Netflix Went Global and to provide great Netflix viewing experience to 190 countries simultaneously. Netflix Built its own globally distributed CDN, **Open Connect**.

Netflix Open Connect delivers 100% of its video traffic, currently over *125 million hours of viewing per day*. This amounts to tens of terabits per second of simultaneous peak traffic, making Netflix Open Connect one of the highest-volume networks in the world.

What CDNs basically do is, they take the original website and the media content it contains, and copy it across hundreds of servers spread all over the world. So when, say, you log in from Mumbai, instead of connecting to the main Netflix server in the United States it will load an exact copy of it from a CDN server that is the closest to Mumbai. This greatly reduces The Latency — the time taken between a request and a response, and everything loads really fast.

But Netflix In Particular takes this Concept a step Further as they directly work with a number of ISPs to install their own hardware called **open connect appliances** at either their own exchange points or even within the ISP's facilities themselves holding up to 280 terabytes of video with the capacity to push out 90 Terabits of data per second, each these come

preloaded with close to the entire Netflix library so what this means for you the consumer is that instead of connecting to some super far away land server to watch a movie you're connecting to an appliance at your own ISP that's much closer cutting down on latency and making it so that your Netflix data packets don't have to fight with all the other internet traffic that is upstream from your ISP and when it's time for catalog updates Netflix pushes them to these appliances.

In a nutshell....

Netflix Has its Own Data Center or CDN's For Media Storage and Transmitting for Everything else, Like Recommender system, user Information, traffic Routing, Log, DNS Routing, etc. Netflix Rely on AWS.

Thanks For Reading 🙏

Connect With Me:

LinkedIn : [Here](#)